

Supplement 1: NAV and ETF Arbitrage

Worked Examples from SEC Filings (BKM Ch. 4)

Big picture: this supplement pairs BKM's conceptual coverage of mutual funds, ETFs, and fund structures with real numbers pulled from SEC EDGAR filings and a from-scratch derivation of why in-kind redemptions escape capital gains tax — so you leave with the formula, a worked example, and the actual statute behind it.

Source context: Companion to the slide deck “NAV and ETF Arbitrage — Ch4 Supplement” (Michigan Tech template). Slide references below are to that deck, not the publisher's chapter lecture slides. BKM section references are to *Investments*, 13th Edition, Chapter 4 (*Mutual Funds and Other Investment Companies*).

What This Supplement Adds

No official numbered learning objectives exist for this material — it is enrichment built on top of BKM Chapter 4, not a textbook section in its own right. By the end of it you should be able to:

- Apply the NAV formula to real fund filings, not textbook placeholders.
 - Explain why ETF prices track NAV as tightly as they do, and work the arbitrage math on both sides — creation and redemption.
 - Identify which market participants can actually run that arbitrage, and which can't.
 - Trace the tax treatment of in-kind redemptions back to its actual statutory source.
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1. Fund Structures & Fees (*Supplement Slide 4*) — BKM §4.2, §4.4

- **Open-end funds** continuously issue/redeem shares at NAV; **closed-end funds** have a fixed share count and trade at a market price that can drift from NAV, with no redemption channel to correct it.
- Three sales-load structures: **no-load** (no sales charge), **front-end load** (charged at purchase), **back-end load / CDSC** (charged at redemption, typically shrinking the longer you hold).

2. Finding Fund Data: SEC EDGAR (*Supplement Slide 5*) — supports BKM §4.1, §4.8

- EDGAR is the SEC's free public filing system — the primary source every other data provider (Morningstar, Bloomberg) ultimately draws from.
- Four-step path: sec.gov/edgar → search by company/fund/ticker → pick the filing type (10-K, N-CSR, N-CSRS) → open it. Every number in the detail sections below came from exactly this path.

3. The Mutual Fund Industry's Growth & Morningstar (*Supplement Slide 6*) — BKM §4.8

- Industry assets grew from under \$1 billion (1945) to over \$200 billion (1984, Morningstar's founding year) to **\$31 trillion+** in mutual funds alone today.
- Morningstar's 1985 star rating and 1992 Style Box gave investors their first standardized way to compare funds — still the industry's standard reference, which is why BKM itself lists [morningstar.com](https://www.morningstar.com) in §4.8.

4. Calculating NAV — Formula & Real Filings (*Supplement Slides 7-10*) — BKM §4.1

- $NAV = (Total\ Assets - Total\ Liabilities) \div Shares\ Outstanding$ — see Key Equations. Published “net assets” figures are already assets minus liabilities; the full gross breakdown only shows up in a fund's SEC shareholder report.
- Summary comparison across three real funds — **VFIAX**, **FCNTX**, **AGTHX** — net assets, NAV, and implied shares outstanding.
- Full Statement of Assets and Liabilities pulled directly from **AGTHX's** and **VFIAX's** actual N-CSR filings, with every line item reconciling to the published NAV.

5. How ETFs Trade (*Supplement Slides 12-14*) — BKM §4.6

- Mutual funds price once a day at NAV (“forward pricing”); ETFs trade continuously all day like a stock — which is also why they carry intraday volatility and bid-ask spread costs mutual fund investors don't.
- **Secondary market** (investors trading existing shares) vs. **primary market** (Authorized Participants creating/redeeming “creation units” directly with the fund) — only the primary market can actually force price back to NAV.
- The full arbitrage mechanism, step by step: secondary-market price drift → AP steps in → creation or redemption → fund issues/cancels creation units.

6. The Arbitrage Mechanism in Numbers (*Supplement Slides 15-16*) — BKM §4.6

- **Worked creation example:** ETF at a \$100.40 premium to \$100 NAV — AP buys the \$5,000,000 basket, delivers it for 50,000 new shares, sells them at market for a **\$20,000 gross profit**; new supply pushes price back down.
- **Worked redemption example:** same ETF at a \$99.60 discount — AP buys 50,000 shares, redeems for the basket, sells the basket for the same **\$20,000 gross profit**; reduced supply pushes price back up.

7. Who Can Arbitrage? APs vs. Non-APs (*Supplement Slide 17*) — BKM §4.6, extension

- Direct creation/redemption is legally restricted to registered **Authorized Participants** — a firm granted that access is an AP by definition.
- Non-AP firms can still arbitrage the secondary market (trade the ETF against the basket, bet on convergence) or route through an actual AP as agent — just with more risk, since nothing forces convergence.

8. Tax Treatment of In-Kind Redemptions (*Supplement Slides 18-19*) — BKM §4.5

- In-kind redemption is why ETFs avoid distributing capital gains the way mutual funds can be forced to when meeting net redemptions.
- The actual mechanism: **IRC §311(b)** is the default rule (distributing appreciated property triggers gain as if sold); **IRC §852(b)(6)** is the specific override for regulated investment companies redeeming stock that's redeemable on demand — which is also exactly why this protection doesn't extend to closed-end funds.

Key Terms

Make sure you can define each of these in your own words.

- ☐ net asset value (NAV)
 - ☐ open-end fund
 - ☐ closed-end fund
 - ☐ no-load fund
 - ☐ front-end load
 - ☐ back-end load (CDSC)
 - ☐ SEC EDGAR
 - ☐ Form N-CSR / N-CSRS
 - ☐ Statement of Assets and Liabilities
 - ☐ forward pricing
 - ☐ Authorized Participant (AP)
 - ☐ creation unit
 - ☐ primary market
 - ☐ secondary market
 - ☐ creation
 - ☐ redemption
 - ☐ in-kind redemption
 - ☐ carryover basis
 - ☐ regulated investment company (RIC)
 - ☐ IRC §311(b)
 - ☐ IRC §852(b)(6)
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Key Equations

Net asset value (NAV)

$$\text{NAV} = \frac{\text{Total Assets} - \text{Total Liabilities}}{\text{Shares Outstanding}}$$

Fact sheets report “net assets” — that's already Assets – Liabilities. The full gross breakdown only appears in a fund's SEC shareholder report (Form N-CSR / N-CSRS).

Gross arbitrage profit (creation / redemption)

$$\text{Gross Profit} = |\text{Market Price} - \text{NAV}| \times \text{Creation Unit Shares}$$

Profit scales with both the size of the mispricing and the creation unit size. In the worked examples, a \$0.40 (0.4%) gap on a 50,000-share unit produced \$20,000 gross profit before trading costs and the fund's own transaction fee.

Study guide for the “NAV and ETF Arbitrage — Ch4 Supplement” slide deck. Slide references are to that deck, not the standard chapter lecture slides. BKM section references are to Investments, 13th Edition, Chapter 4 (Mutual Funds and Other Investment Companies).